

Mudit Garg

📧: [astromuditgarg.github.io](https://github.com/astromuditgarg)

✉: mg9113@nyu.edu

Research: [ADS Library](#)

EDUCATION

08/'21 – 10/'25	Doctor of Philosophy in Gravitational Wave Astrophysics University of Zurich <i>Decoding the Sub-Parsec Scale Environment of LISA Massive Black Hole Binaries with Gravitational Waves</i>	Advisor: Prof. Lucio Mayer
09/'18 – 12/'20	Master of Science in Physics with distinction ETH Zurich <i>Relativistic, ghost-free, and covariant hybrid model for MOND: $f(Q)$</i>	GPA: 5.87/6 Thesis supervisor: Prof. Lavinia Heisenberg
07/'14 – 06/'18	Bachelor of Technology in Engineering Physics IIT Delhi <i>Geodesics near a charged black hole in $(R \pm \mu^4/R)$ gravity</i>	GPA: 8.15/10 Thesis supervisor: Prof. Ajit Kumar

ACADEMIC EXPERIENCE

11/'25 – 10/'27	Swiss Mobility Postdoctoral Fellow in Astrophysics New York University	Host: Prof. Andrew MacFadyen
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SELECTED TALKS - 6 SEMINARS, 4 INVITED + 8 CONTRIBUTED CONFERENCES, AND 7 INDIVIDUAL

10/'24	CTC Theory Seminar at University of Maryland [25+25 minutes] <i>Decoding Astrophysics from inspiraling LISA MBHBs</i>	College Park
09/'24	Astrophysics Seminar at Johns Hopkins University [45+15 minutes] <i>Decoding Astrophysics from inspiraling LISA MBHBs</i>	Baltimore
09/'24	Astro Seminar at Columbia University <i>Decoding Astrophysics from inspiraling LISA MBHBs</i>	NYC
06/'24	GRAPPA Seminar at University of Amsterdam [45+15 minutes] <i>Astrophysical signatures on the LISA data stream from MBHBs</i>	Amsterdam
05/'24	Cosmology Seminar at Max Planck Institute for Astrophysics <i>Astrophysical signatures on GWs from LISA MBHBs</i>	Garching
02/'24	DAMTP GR Seminar at University of Cambridge [50+10 minutes] <i>Astrophysical signatures on the LISA data stream from MBHBs</i>	Cambridge
06/'26	16th LISA Symposium at University of Maryland <i>Chaotic migration of LISA EMRIs in a turbulent accretion disk: effect on waveform de-phasing</i>	College Park
06/'25	Workshop: Astrophysical Dynamics: from planets, to stars, to black holes Niels Bohr Institute <i>Characterizing sub-pc environment of LISA MBHBs</i>	Copenhagen
06/'25	Conference (Invited): DYNAMIX Institute of Astronomy, Cambridge <i>Characterizing sub-pc environment of LISA MBHBs</i>	Cambridge
05/'25	Workshop (Invited): Gravitational Wave Probes of Black Hole Environments IFPU, SISSA & ICTP <i>Characterizing sub-pc environment of LISA MBHBs</i>	Trieste
03/'25	Workshop (Invited): Frontiers of Astrophysical Black Holes Sexten Center for Astrophysics <i>What solves the 'final parsec' problem for LISA MBHBs?</i>	Sexten
08/'24	Conference (Invited): New ideas on the origin of BH mergers Niels Bohr Institute <i>Astrophysical signatures on the LISA data stream from MBHBs</i>	Copenhagen
11/'24	Meeting: LISA Astrophysics Working Group at MPA <i>What solves the 'final parsec' problem for LISA Massive Black Hole Binaries?</i>	Garching

09/'23	Meeting: LISA Astrophysics Working Group at University of Milano-Bicocca <i>The minimum measurable eccentricity from GWs of LISA MBHBs</i>	Milan
07/'23	Conference: GW populations: what's next? at University of Milano-Bicocca <i>The measurability of gas and eccentricity from GWs of LISA MBHBs</i>	Milan
11/'22	Conference: LISA data analysis: classical methods to machine learning CNRS, L2IT, APC, CEA, and CNES <i>The imprint of Gas on GWs from LISA IMBHBs</i>	Toulouse
09/'22	Conference: Origin, growth and feedback of BHs in dwarf galaxies Donostia International Physics Center <i>The imprint of Gas on GWs from LISA IMBHBs</i>	San Sebastian
05/'22	Conference: IMBHBs: New Science from Stellar Evolution to Cosmology CIERA, Northwestern University <i>Gas impact on GWs from LISA IMBHBs</i>	San Juan
10/'24	CIERA theory group meeting at Northwestern University <i>Decoding Astrophysics from inspiraling LISA MBHBs</i>	Evanston
09/'24	Branch Lunch at NASA Goddard <i>Decoding Astrophysics from inspiraling LISA MBHBs</i>	Greenbelt
09/'24	Astro Coffee at Institute of Advanced Study <i>Measuring eccentricity and gas-induced perturbation from GWs of LISA MBHBs</i>	Princeton
09/'24	Monday Afternoon Talks at MIT Kavli Institute <i>Decoding Astrophysics from inspiraling LISA MBHBs</i>	Boston
07/'24	15th LISA Symposium at University College Dublin <i>Poster: Astrophysical signatures on the LISA data stream from MBHBs</i>	Dublin
02/'25	LISA Call Systematics in tests of GR using LISA MBHBs (invited)	Online
06/'24	Measuring eccentricity and gas from GWs of LISA MBHBs	Community
07/'23	Measuring eccentricity from GWs of LISA MBHBs	Data Challenge Working Group

RESEARCH VISITS

06/'25	Niels Bohr International Academy, University of Copenhagen <i>Host: Prof. Johan Samsing</i>	Copenhagen
10/'24	Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA) <i>Host: Prof. Shane Larson</i>	Evanston
09/'24	Center for Computational Astrophysics (CCA), Flatiron Institute <i>Host: Prof. Will Farr, Dr. Yan-Fei Jiang, and Dr. Matteo Cantiello</i>	NYC
02/'24	Institute of Gravitational Wave Astronomy <i>Host: Prof. Alberto Vecchio</i>	Birmingham
02/'24	Institute of Cosmology and Gravitation <i>Host: Prof. Ian Harry</i>	Portsmouth
11/'23	Max Planck Institute for Gravitational Physics (Albert Einstein Institute) <i>Host: Dr. Jonathan Gair</i>	Potsdam

PROGRAMS/SCHOOLS

07/'25	MIAPbP Program: Enabling Future GW Astrophysics in the Milli-Hertz Regime <i>Munich Institute for Astro-, Particle and BioPhysics</i>	Garching
09/'24	Workshop: Fundamental Physics Meets Waveforms With LISA <i>Max Planck Institute for Gravitational Physics (Albert Einstein Institute)</i>	Potsdam
09/'23	Kavli-Villum School: Gravitational Waves <i>Corfu Summer Institute</i>	Corfu

11/'22	Workshop: LISA data analysis: classical methods to machine learning <i>CNRS, L2IT, APC, CEA, and CNES</i>	Toulouse
07/'22	Workshop: LISA Data Challenge Workshop <i>LISA Data Challenge Working Group</i>	Online
01/'22	Saas-Fee School: Multi-Messenger GW Astronomy <i>Swiss Society for Astrophysics and Astronomy</i>	Saas-Fee
08/'21	NBIA School: Gravitational wave astrophysics <i>Niels Bohr Institute, University of Copenhagen</i>	Copenhagen

PROFESSIONAL RESPONSIBILITIES AND MEMBERSHIPS

'25 –	Referee for ApJ, PRD
'23 – '25	Organizer of the ‘GWs, BHs, and Compact Binaries’ seminar <i>Department of Astrophysics, University of Zurich</i>
'22 – '26	Contributor to the LISA DiscIMRI hydrodynamical code comparison project <i>Tasks: Literature review, plot making, and text writing</i>
'21 –	Member of the LISA consortium, its astrophysics, waveforms, and data challenge working groups, and its early career scientist group (LECS)
'21 – '25	Teaching assistant for several astrophysics courses at the University of Zurich
'21	Research assistant at the Department of Astrophysics, University of Zurich

SKILLS

Software and programming language

- **GIZMO:** Performed and analyzed simulations of gravitational-wave driven LISA massive black hole binaries embedded in accretion disk
- **LISABETA:** Added several waveform modules. Also, experienced user and frequently edits it to suit a given project’s needs. It provides a complete LISA response and Bayesian inference primarily using the PTMCMC sampler.
- **ATHENA++:** Learning how to simulate LISA binaries embedded in an accretion disk
- **ERYN:** I have performed reversible jump MCMC with this sampler in LISABETA.
- **MATHEMATICA:** Frequent user to do analysis and plotting.
- **PYTHON:** I mainly use this programming language to perform analysis and make plots.

Languages: English | German (A1.1) | Hindi

Last update: May 12, 2026